

Project 10: Polyglot Monorepo CI/CD Demo

Overview

This project demonstrates how to structure and deploy a **Monorepo** containing multiple applications written in different languages. The key learning objective is **Pipeline routing**: building a Java backend and Python frontend in one Jenkinsfile.

We will build two separate Docker images (`polyglot-backend` and `polyglot-frontend`) and deploy them to the shared `jenkins-net` network, where the frontend will communicate with the backend via internal DNS resolution.

Prerequisites

Ensure the following are installed and running on your Jenkins server:

- Docker & Trivy
- Python 3 & pip
- Java 17 & Maven

Step 1: Network Setup

Create a shared Docker network for your infrastructure and deployments:

```
docker network create jenkins-net
```

Step 2: Infrastructure Setup

Run SonarQube and Jenkins on the `jenkins-net` network:

```
# Run SonarQube
```

```
docker run -d --name sonarqube -p 9000:9000 --network jenkins-net  
sonarqube:lts-community
```

```
# Run Jenkins
```

```
docker run -d --name jenkins -p 8081:8080 -p 50000:50000 --network  
jenkins-net -v /var/run/docker.sock:/var/run/docker.sock -v  
jenkins_home:/var/jenkins_home jenkins/jenkins:lts
```

(Map Jenkins to 8081 if 8080 is used by the backend app).

Step 3: Jenkins Configuration

1. Log into Jenkins.
2. Install necessary plugins: Docker Pipeline, SonarQube Scanner, Pipeline.
3. Configure SonarQube Server in Manage Jenkins -> System.
4. Configure Global Tools (Maven, JDK, Sonar Scanner).

Step 4: Pipeline Execution

1. Create a new Pipeline job in Jenkins.
2. Choose "Pipeline script from SCM" or paste the contents of Jenkinsfile.
3. Run the build. Observe how Jenkins navigates into the `backend/` and `frontend/` directories using the `dir()` step to execute language-specific builds!

Step 5: Verification

Access the deployed Frontend Application, which will securely query the Java Backend:

- URL: `http://localhost:5000`

- Expected Output: `{ "backend\_response": { "message": "Hello from the Java Spring Boot Backend!" }, "frontend\_status": "Active" }`